



SERVING UP A VIRTUALISED WORLD

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A business, be it small, medium sized or large, has some basic computing prerequisites. First there should be IT stability. This means that downtimes are strictly unacceptable. Productivity should ideally never have to be halted on account of failure in the IT backbone of a company. Another prerequisite that makes complete business sense is lowering the cost of infrastructure. Most businesses these days are trying to lower their TCO (Total Cost of Ownership) by any means possible.

A buzzword that's been doing the rounds lately promises to take care of the above requirements. That word is virtualization. This solution is expected to improve utilisation, reduce costs,

Using virtualization to service your business needs

increase productivity and aid disaster recovery. The current adoption rate of virtualization in India is at around 19 per cent but is expected to quadruple in the next two years.

What is Virtualization?

Virtualization basically means abstraction of computer resources. But to get a better understanding of what that really means, one must delve deeper into its forms. Virtualization has several variants but the ones that makes most sense in the business arena are platform virtualization, and — at the other end of the spectrum — desktop virtualization.

Let's look at platform virtualization first. Platform virtualization is performed on a given hardware platform by virtualization software, which creates a simulated computer environment, a virtual machine, for its guest software. The guest software here is a fully functional operating system. This means that one piece of hardware or computer can have several virtual machines (VMs) running within it simultaneously. Each one of those machines has its own virtual components such as memory and storage space. Each one of the VMs have an operating system. This is all possible thanks to advances in hardware technology such as quad core processors that are so powerful that they have plenty of resources to spare. In such a scenario, you actually have one server but it acts as though it were several servers. Virtualization software enables that server to be split up into different partitions. With virtualization, each server's programs, files, and processes are merged into one image file. This will be one of the many image files constituting each of the virtual servers.

Therefore, one physical server could act as, say, three virtual servers with each virtual server performing a specific, and different, function. You could have a Windows-based file server, while another could be a Linux web server hosting web sites on Apache. Another could be a DBMS server based on another platform. Each virtual server acts completely independently from one another, hosted on a virtual abstraction layer provided by the virtualization software. If one server crashes, the others are not affected. The net result is that you have to buy only one powerful physical server and you get the benefit of three servers for the cost of that one.

Desktop virtualization, on the other hand, is a variant of the thin client model, only with greater flexibility. For example, users can use a stripped down PC as a client, connect to virtual resources and software on a server, but at the same time have the freedom to install and configure their own applications. A user's server-based desktop can be accessed from other locations as well. However, since this variant is discussed more commonly, we will look more at hardware or server virtualization.

What server virtualization software is available and how does it work?

VMware currently has the largest market share worldwide when it comes to server virtualization software. It has both free as well as paid options. VMware Server and VMware ESXi are free solutions. Virtualization software is installed as an application on top of a host Windows or Linux operating system. Once installed, a thin virtualization layer partitions the physical server to be able to run multiple VMs simultaneously. The computing resources of the physical server are pooled and allocated to the VMs as per their needs. The layer of virtualization ensures that each VM is isolated from its host and other VMs. Within the limits of your hardware configuration you generally get near native performance.

The other options for server virtualization software are paid solutions such as XenServer from Citrix and Parallel's Virtuozzo Containers. In the open source world you have OpenVZ and Linux V-Server. There is also a popular application called Virtual Box, which many home users would've tried.



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